RAVI KUMAR NERSU

Senior Frontend / Full-Stack Engineer | React.js | Next.js | TypeScript | Node.js / NestJS | AI-Assisted Engineering

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PROFESSIONAL SUMMARY

Senior Frontend / Full-Stack Engineer with 8.6 years of experience building production-grade web platforms using React.js, Next.js, TypeScript, JavaScript and Node.js / NestJS. Owned Avis Budget Group frontend surfaces across 15+ country markets, supporting customer-facing digital experiences used by millions of global users. Built NestJS BFF services with clean Controller → Service → Repository architecture, JWT/OAuth2 authentication and role-based access control (RBAC), PostgreSQL/Prisma and Redis-backed data flows, Docker-based containerized delivery and GitHub Actions CI/CD. Improved Lighthouse score from ~45 to ~80, page load from ~6s to ~3s and LCP from ~4.5s to ~2.2s through SSR/SSG decisions, route-level code splitting, image optimization and JS payload reduction. Experienced in Micro Frontends (MFEs), BFF services, AEM Headless CMS, REST/GraphQL API contracts, Jest, React Testing Library, integration testing, CI/CD release delivery, PR reviews and mentoring in product/GCC engineering teams. Hands-on with AI-assisted software engineering using Claude Code, Claude for VS Code and GitHub Copilot for implementation planning, debugging, refactoring, test ideation and documentation.

KEY ACHIEVEMENTS

- **Performance impact:** improved Lighthouse score ~45 → ~80, page load ~6s → ~3s, LCP ~4.5s → ~2.2s, CLS ~0.25 → ~0.05 and TBT ~600ms → ~250ms in optimization checks.
- **Secure full-stack delivery:** implemented JWT/OAuth2 authentication and RBAC across the Next.js frontend and NestJS BFF, with PostgreSQL/Prisma data access, Redis caching and Docker-based containerized deployment.
- **Scale and ownership:** owned multi-market booking/search, forms, dashboards and AEM-driven header/navigation surfaces across 15+ country markets for Avis digital platforms.
- **Frontend efficiency:** reduced initial JS payload from ~1.2 MB to ~800 KB using route-level code splitting, bundle analysis, image optimization and rendering improvements.
- **Quality ownership:** used Jest, React Testing Library and integration testing practices for React/Next.js features, reusable components and critical booking-related flows.
- **AI-assisted engineering:** used Claude Code, Claude for VS Code and GitHub Copilot for development acceleration, refactoring, debugging, test-case ideation and documentation; completed GenAI certification.

CORE SKILLS

Frontend	React.js, Next.js App Router, TypeScript, JavaScript ES6+, Redux, Zustand, React Query, HTML5, CSS3, Tailwind CSS, Shadcn UI
Architecture	Server Side Rendering (SSR), SSR/SSG, Micro Frontends (MFEs), Frontend Architecture, Design Systems, Component Systems, BFF Pattern, Clean Architecture (Controller → Service → Repository), API Contracts, Core Web Vitals, Performance Budgets, Accessibility, SEO
Backend & APIs	Node.js, NestJS, Express.js, REST APIs, GraphQL, JWT/OAuth2 Authentication, Role-Based Access Control (RBAC), Microservices Integration, Secure Data Flows
Data & Search	PostgreSQL, MySQL, MongoDB, Redis, Prisma, Typesense, Indexing, Query Optimization, Transactions, Pagination, Filtering, Data Modelling
DevOps & Quality	Git, GitHub Actions, Docker, Containerized Deployment, CI/CD, AWS EC2/S3/CloudFront, Webpack, Vite, npm/Yarn/pnpm, Jest, React Testing Library, Unit Testing, Integration Testing, Vitest, Playwright, Jira, Artifactory
AI-Assisted Engineering	Claude Code, Claude for VS Code, GitHub Copilot, GenAI Certification, Prompt Engineering Basics, AI-assisted Development, AI-assisted Code Review, Test Ideation, Documentation, Claude Code Workflows, Subagents, Agent Skills, BMAD Method, Backlog.md

PROFESSIONAL EXPERIENCE

LTIMindtree | Hyderabad, India **Jan 2024 – Apr 2026**

Specialist Software Engineer – Frontend / Full-Stack Engineering | Client: Avis Budget Group

- Owned frontend architecture for Avis digital platform surfaces across 15+ country markets, defining rendering strategy, route structure, API boundaries, reusable UI patterns and caching approach with React, Next.js App Router and Node.js/NestJS BFF services using clean Controller → Service → Repository architecture.
- Implemented Server Side Rendering (SSR), SSG and route-level rendering decisions for customer-facing pages to improve SEO, load performance and Core Web Vitals across multi-market web journeys.
- Implemented JWT/OAuth2-based authentication and role-based access control (RBAC) across the Next.js frontend and NestJS BFF — covering token validation, protected routes, refresh/session handling and role-gated UI for secure multi-market customer journeys.
- Designed REST/GraphQL API integrations through BFF services with validation flows, error states, loading states, secure data handling, API contract alignment and graceful failure behavior for production web journeys.
- Integrated Redis for in-memory caching and session management and used MongoDB for production data access patterns and API-backed workflows, alongside relational data integrations, to improve data-flow performance.
- Improved Lighthouse performance score from ~45 to ~80, page load from ~6s to ~3s and LCP from ~4.5s to ~2.2s by applying SSR/SSG tradeoffs, Redis/MongoDB-backed data flows, route-level code splitting, image optimization and bundle analysis.
- Reduced initial JS payload from ~1.2 MB to ~800 KB and improved Core Web Vitals signals including CLS ~0.25 → ~0.05 and TBT ~600ms → ~250ms based on Lighthouse/network analysis.
- Worked on Micro Frontend (MFE) patterns for modular frontend surfaces, reusable market-specific UI modules and independently maintainable React/Next.js feature areas.
- Built React/TypeScript component modules for booking-related flows, search/filter experiences, reusable forms, dashboards, CMS-driven UI experiences, responsive layouts and accessible interaction patterns.
- Wrote and maintained unit tests and integration tests using Jest and React Testing Library for reusable components, form states, API-driven UI flows and critical booking-related interactions.
- Integrated AEM Headless CMS for reusable header/navigation, market-specific content and CMS-driven UI surfaces; coordinated content models, API contracts and release dependencies with enterprise platform teams.
- Containerized services with Docker and shipped through GitHub Actions CI/CD — supported multiple QA/UAT/staging deployments per week and one production release every 2-week sprint using Webpack, Vite, Yarn and Artifactory.
- Leveraged Claude Code, Claude for VS Code and GitHub Copilot for implementation planning, debugging, refactoring, unit-test ideation, technical documentation and code review preparation, with manual review of generated suggestions.
- Worked in a 6-engineer team, mentored 2 developers, reviewed PRs for 3 developers, and coordinated with product, backend, QA, DevOps and CMS teams for multi-market releases.

Tech: React, Next.js, TypeScript, JavaScript, Node.js, NestJS, JWT/OAuth2, RBAC, AEM Headless CMS, REST APIs, GraphQL, MongoDB, Redis, Tailwind CSS, Shadcn UI, Micro Frontends, Jest, React Testing Library, Integration Testing, Docker, AWS, GitHub Actions, Webpack, Vite, Yarn, Artifactory, Jira, Claude Code, Claude for VS Code, GitHub Copilot

MSR Cosmos IT LLP | Hyderabad, India **Aug 2020 – Jan 2024**

Senior Software Engineer | Client: TMILL / Tata NYK Logistics

- Owned end-to-end frontend modules for logistics automation across Railways, SCM and MLS domains, translating operational workflows into production web applications using React, Next.js and Node.js.
- Built data-intensive dashboards for live tracking, geospatial visualization and operational reporting using React, Next.js, Node.js, PostgreSQL and Google Maps API.
- Designed and integrated backend services, REST APIs and PostgreSQL-backed workflows for high-volume logistics operations, including schema design, complex queries, indexing, transactions and query optimization.
- Improved workflow performance and maintainability through frontend optimization, reusable components, query tuning, accessibility improvements and production debugging.
- Worked directly with business stakeholders to clarify requirements, estimate delivery scope and convert domain workflows into scalable technical solutions.

Tech: React, Next.js, JavaScript, Node.js, REST APIs, PostgreSQL, MySQL, Google Maps API

Allmvr Projects Pvt. Ltd | Vijayawada, India **Jun 2019 – Aug 2020**

Front-End Developer

- Built customer-facing and admin web applications for a multi-vendor retail delivery platform, including onboarding, ordering, payments, dashboards and real-time location tracking flows.
- Converted Figma designs into responsive React interfaces and integrated REST APIs for customer and admin workflows.

Tech: React, JavaScript, HTML5, CSS3, REST APIs

Xelpro Software Solutions & Whatsscore Pvt. Ltd | Hyderabad, India **Jun 2017 – Jun 2019**

Software Developer

- Developed web and mobile application modules for sports tournament and event management platforms, including authentication, registration, dashboards and reusable UI modules.
- Supported debugging, maintenance and feature enhancements across multiple client-facing applications.

Tech: JavaScript, React, HTML5, CSS3, REST APIs

SELECTED PROJECT / HANDS-ON POC

AI Hub Platform POC – Demand Intake, Governance and Discovery | Full-Stack React/Node Enterprise Workflow POC

- Built a full-stack POC with demand intake forms, governance dashboards, AI use-case catalogue, role-based access (RBAC) screens and approval workflow patterns.
- Implemented the POC using React, TypeScript, Node.js APIs, Prisma/PostgreSQL, Tailwind CSS and Shadcn UI; explored Wasp, Typesense and enterprise API integration patterns.
- Used Claude Code, Claude for VS Code and GitHub Copilot for code exploration, refactoring support, documentation and test-case ideation, with manual review of generated code.
- Completed GenAI certification and followed a structured AI-assisted engineering upskilling plan covering Claude Code workflows, subagents, agent skills, BMAD Method, Backlog.md, Node/git/local setup and AI coding workflow practices.

Tech: React, TypeScript, Node.js, Wasp, Prisma, PostgreSQL, Tailwind CSS, Shadcn UI, Typesense, GitHub Actions, Claude Code, Claude for VS Code, GitHub Copilot

EDUCATION

KL University, Guntur, India – B.Tech, Computer Science & Engineering | Jun 2013 – May 2017